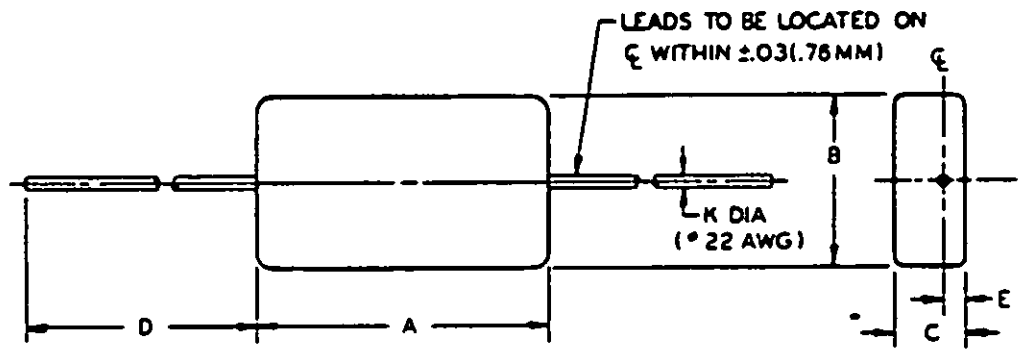
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC,

STYLE CM20

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.



DIM	DIMENSIONS					
	INCHES			MILLIMETERS		
	MIN	NOM	MAX	MIN	NOM	MAX
A	.67	.73	.79	17.02	18.54	20.07
B	.41	.44	.47	10.41	11.18	11.94
C	.16	.19	.22	4.06	4.83	5.59
D	1.12	---	---	28.45	---	---
E	.06	---	---	1.52	---	---
K	.030	.032	.034	.76	.81	.86

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Approximate weight .10 ounce.
4. Unless otherwise specified, tolerance is .xx ±.02.

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REQUIREMENTS:

Dimensions: See figure.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Symbol	Capacitance tolerance
F	± 1 percent
G	± 2 percent
J	± 5 percent

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55° C to +85° C
O	-55° C to +125° C
P	-55° C to +150° C

Vibration grade: 3 (10 to 2,000 cps).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/°C	Capacitance drift
C	-200 to +200	±(0.5 percent + 0.1 pf)
E	-30 to +100	±(0.1 percent + 0.1 pf)
F	0 to +70	±(0.05 percent + 0.1 pf)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202: Condition D (100,000 ft.)
Test potential - 150 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25° C - 100,000 megohms minimum.
At 85° C - 15,000 megohms minimum.
At 125° C - 10,000 megohms minimum.
At 150° C - 5,000 megohms minimum.

Dissipation factor, maximum: See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Vibration:
Method 204 of MIL-STD-202, condition D.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except the high test temperature shall be as follows:
Symbol O - +125°C.
Symbol P - +150°C.

Immersion cycling:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Shock (medium impact):
Method 205 of MIL-STD-202, condition C (50G).

Solderability:
Method 208 of MIL-STD-202.

Terminal strength:
Pull test - Capacitor held by one terminal and 5 pound load gradually applied to other terminal and maintained for at least 5 seconds.
Twist test - 5 alternate 360° rotations.

Moisture resistance:
Method 106 of MIL-STD-202.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

Temperature coefficient and capacitance drift:
Within the limits specified for the applicable characteristic.

Low temperature storage and life:
48 hours at -55 $^{+0}_{-2}$ ° C, immediately followed by qualification or performance test, as applicable.
Qualification test - 2,000 hours at applicable high test temperature with 150 percent of rated dc working voltage applied.
Dielectric withstanding voltage - Initial requirement.
Insulation resistance - Initial requirement.
Dissipation factor - Not greater than 150 percent of initial requirement.
Capacitance change - Shall not exceed ±2 percent of nominal value or 1 pf, whichever is greater.

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Type 1/ designation
CM20-D561--3
CM20-D621--3
CM20-D681--3
CM20-D751--3
CM20-D821--3
CM20-D911--3
CM20-D102--3
CM20-D112--3
CM20-D122--3
CM20-D132--3
CM20-D152--3
CM20-D162--3
CM20-D182--3
CM20-C202--3
CM20-C212--3
CM20-C242--3
CM20-C272--3
CM20-C302--3

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TABLE I. Style CM20

Capacitance	DC working voltage	Operating temperature range	Characteristics available for capacitance tolerance		
			±1% (F)	±2% (G)	±5% (J)
(pf)	(volts)				
560	500	N, O, P	CEF	CEF	CEF
620	500	N, O, P	CEF	CEF	CEF
680	500	N, O, P	CEF	CEF	CEF
750	500	N, O, P	CEF	CEF	CEF
820	500	N, O, P	CEF	CEF	CEF
910	500	N, O, P	CEF	CEF	CEF
1,000	500	N, O, P	CEF	CEF	CEF
1,100	500	N, O	CEF	CEF	CEF
1,200	500	N, O	CEF	CEF	CEF
1,300	500	N, O	CEF	CEF	CEF
1,500	500	N, O	CEF	CEF	CEF
1,600	500	N, O	CEF	CEF	CEF
1,800	500	N, O	CEF	CEF	CEF
2,000	300	N, O	CEF	CEF	CEF
2,200	300	N, O	CEF	CEF	CEF
2,400	300	N, O	CEF	CEF	CEF
2,700	300	N, O	CEF	CEF	CEF
3,000	300	N, O	CEF	CEF	CEF

plete type designation will include additional symbols to indicate
racteristic, capacitance tolerance and temperature range.

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Performance check - Conditions and requirements in accordance with qualification test except test duration shall be 250 hours.

Continuation test - Units that have completed the 250 hour performance shall be continued for an additional 1,750 hours and meet the requirements for the qualification test.

Qualification inspection: Qualification inspection (Group A) is required.

Capacitors as described herein with capacitance values ranging from 560 through 1,000 pf, may be considered as suitable replacements for comparable items formally covered by MIL-C-5/3C. Recommended replacements are as follows:

MIL-C-5/3C items	Recommended military replacement items
CM25B561JN3 thru CM25B102JN3	CM20CD561JN3 thru CM20CD102JN3
CM25B561JO3 thru CM25B102JO3	CM20CD561JO3 thru CM20CD102JO3
CM25B561JP3 thru CM25B102JP3	CM20CD561GP3 thru CM20CD102JP3
CM25E561GN3 thru CM25E102GN3	CM20ED561GN3 thru CM20ED102GN3
CM25E561GO3 thru CM25E102GO3	CM20ED561GO3 thru CM20ED102GO3
CM25E561GP3 thru CM25E102GP3	CM20ED561GP3 thru CM20ED102GP3
CM25E561JN3 thru CM25E102JN3	CM20ED561JN3 thru CM20ED102JN3
CM25E561JO3 thru CM25E102JO3	CM20ED561JO3 thru CM20ED102JO3
CM25E561JP3 thru CM25E102JP3	CM20ED561JP3 thru CM20ED102JP3
CM25F561GN3 thru CM25F102GN3	CM20FD561GN3 thru CM20FD102GN3
CM25F561GO3 thru CM25F102GO3	CM20FD561GO3 thru CM20FD102GO3
CM25F561GP3 thru CM25F102GP3	CM20FD561GP3 thru CM20FD102GP3
CM25F561JN3 thru CM25F102JN3	CM20FD561JN3 thru CM20FD102JN3
CM25F561JO3 thru CM25F102JO3	CM20FD561JO3 thru CM20FD102JO3
CM25F561JP3 thru CM25F102JP3	CM20FD561JP3 thru CM20FD102JP3

Supersession data: Capacitors with voltage ratings of 1,000 volts and greater, with part numbers M5/02-0211 thru -1929, formerly covered by this specification sheet are no longer authorized. Since there has never been a qualified source for these items, no replacement data is offered. For capacitors with voltage ratings of 500 volts and less, see table II.

TABLE II. Part number to replacement type designation.

Dash No.	Replacement item	Dash No.	Replacement item	Dash No.	Replacement item
0001	CM20CD561FN3	0015	CM20ED561GO3	0029	CM20FD561JP3
0002	CM20CD561GN3	0016	CM20ED561JO3	0031	CM20CD621FN3
0003	CM20CD561JN3	0017	CM20FD561FO3	0032	CM20CD621GN3
0004	CM20ED561FN3	0018	CM20FD561GO3	0033	CM20CD621JN3
0005	CM20ED561GN3	0019	CM20FD561JO3	0034	CM20ED621FN3
0006	CM20ED561JN3	0021	CM20CD561FP3	0035	CM20ED621GN3
0007	CM20FD561FN3	0022	CM20CD561GP3	0036	CM20ED621JN3
0008	CM20FD561GN3	0023	CM20CD561JP3	0037	CM20FD621FN3
0009	CM20FD561JN3	0024	CM20ED561FP3	0038	CM20FD621GN3
0011	CM20CD561FO3	0025	CM20ED561GP3	0039	CM20FD621JN3
0012	CM20CD561GO3	0026	CM20ED561JP3	0041	CM20CD621FO3
0013	CM20CD561JO3	0027	CM20FD561FP3	0042	CM20CD621GO3
0014	CM20ED561FO3	0028	CM20FD561GP3	0043	CM20CD621JO3

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TABLE II. Part number to replacement type designation. - Continued

Dash No.	Replacement item	Dash No.	Replacement item	Dash No.	Replacement item
0044	CM20ED621FO3	0099	CM20FD751JN3	0155	CM20ED911GN3
0045	CM20ED621GO3	0101	CM20CD751FO3	0156	CM20ED911JN3
0046	CM20ED621JO3	0102	CM20CD751GO3	0157	CM20FD911FN3
0047	CM20FD621FO3	0103	CM20CD751JO3	0158	CM20FD911GN3
0048	CM20FD621GO3	0104	CM20ED751FO3	0159	CM20FD911JN3
0049	CM20FD621JO3	0105	CM20ED751GO3	0161	CM20CD911FO3
0051	CM20CD621FP3	0106	CM20ED751JO3	0162	CM20CD911GO3
0052	CM20CD621GP3	0107	CM20FD751FO3	0163	CM20CD911JO3
0053	CM20CD621JP3	0108	CM20FD751GO3	0164	CM20ED911FO3
0054	CM20ED621FP3	0109	CM20FD751JO3	0165	CM20ED911GO3
0055	CM20ED621GP3	0111	CM20CD751FP3	0166	CM20ED911JO3
0056	CM20ED621JP3	0112	CM20CD751GP3	0167	CM20FD911FO3
0057	CM20FD621FP3	0113	CM20CD751JP3	0168	CM20FD911GO3
0058	CM20FD621GP3	0114	CM20ED751FP3	0169	CM20FD911JO3
0059	CM20FD621JP3	0115	CM20ED751GP3	0171	CM20CD911FP3
0061	CM20CD681FN3	0116	CM20ED751JP3	0172	CM20CD911GP3
0062	CM20CD681GN3	0117	CM20FD751FP3	0173	CM20CD911JP3
0063	CM20CD681JN3	0118	CM20FD751GP3	0174	CM20ED911FP3
0064	CM20ED681FN3	0119	CM20FD751JP3	0175	CM20ED911GP3
0065	CM20ED681GN3	0121	CM20CD821FN3	0176	CM20ED911JP3
0066	CM20ED681JN3	0122	CM20CD821GN3	0177	CM20FD911FP3
0067	CM20FD681FN3	0123	CM20CD821JN3	0178	CM20FD911GP3
0068	CM20FD681GN3	0124	CM20ED821FN3	0179	CM20FD911JP3
0069	CM20FD681JN3	0125	CM20ED821GN3	0181	CM20CD102FN3
0071	CM20CD681FO3	0126	CM20ED821JN3	0182	CM20CD102GN3
0072	CM20CD681GO3	0127	CM20FD821FN3	0183	CM20CD102JN3
0073	CM20CD681JO3	0128	CM20FD821GN3	0184	CM20ED102FN3
0074	CM20ED681FO3	0129	CM20FD821JN3	0185	CM20ED102GN3
0075	CM20ED681GO3	0131	CM20CD821FO3	0186	CM20ED102JN3
0076	CM20ED681JO3	0132	CM20CD821GO3	0187	CM20FD102FN3
0077	CM20FD681FO3	0133	CM20CD821JO3	0188	CM20FD102GN3
0078	CM20FD681GO3	0134	CM20ED821FO3	0189	CM20FD102JN3
0079	CM20FD681JO3	0135	CM20ED821GO3	0191	CM20CD102FO3
0081	CM20CD681FP3	0136	CM20ED821JO3	0192	CM20CD102GO3
0082	CM20CD681GP3	0137	CM20FD821FO3	0193	CM20CD102JO3
0083	CM20CD681JP3	0138	CM20FD821GO3	0194	CM20ED102FO3
0084	CM20ED681FP3	0139	CM20FD821JO3	0195	CM20ED102GO3
0085	CM20ED681GP3	0141	CM20CD821FP3	0196	CM20ED102JO3
0086	CM20ED681JP3	0142	CM20CD821GP3	0197	CM20FD102FO3
0087	CM20FD681FP3	0143	CM20CD821JP3	0198	CM20FD102GO3
0088	CM20FD681GP3	0144	CM20ED821FP3	0199	CM20FD102JO3
0089	CM20FD681JP3	0145	CM20ED821GP3	0201	CM20CD102FP3
0091	CM20CD751FN3	0146	CM20ED821JP3	0202	CM20CD102GP3
0092	CM20CD751GN3	0147	CM20FD821FP3	0203	CM20CD102JP3
0093	CM20CD751JN3	0148	CM20FD821GP3	0204	CM20ED102FP3
0094	CM20ED751FN3	0149	CM20FD821JP3	0205	CM20ED102GP3
0095	CM20ED751GN3	0151	CM20CD911FN3	0206	CM20ED102JP3
0096	CM20ED751JN3	0152	CM20CD911GN3	0207	CM20FD102FP3
0097	CM20FD751FN3	0153	CM20CD911JN3	0208	CM20FD102GP3
0098	CM20FD751GN3	0154	CM20ED911FN3	0209	CM20FD102JP3

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Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL
Navy - SH, AS, OS, EC
Air Force - 11, 17, 85

User activities:
Army -
Navy - MC, CG
Air Force - 14, 19

Preparing activity:
Navy - SH

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